Edexcel A Level Business

Your notes

1.2 Market

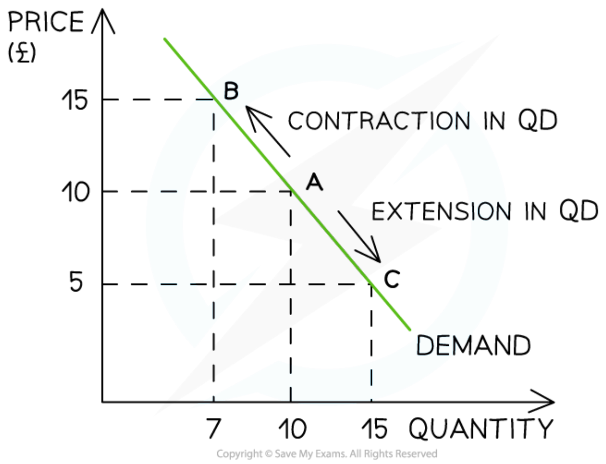
Contents

Your notes

An introduction to demand

- Demand refers to the **number of goods/services customers are willing to buy** at a given price
  - **Effective demand** occurs when customers are **willing and able** (they have the money) to buy at a given price
- There is an **inverse relationship** between the quantity demanded by customers and the price
  - As the price increases, the quantity demanded decreases
  - As the **price decreases, the quantity demanded increases**
  - Hence, the **demand curve slopes downwards** from left to right
- This is illustrated in the diagram below

A simple demand curve



A demand curve showing how a change in price will lead to a change in quantity demanded (QD)

Diagram analysis

- An **increase** in price from £10 to £15 leads to a movement **up** the demand curve from point A to point B
  - Due to the **increase** in price, the quantity demanded (QD) has **fallen** from 10 to 7 units
- A **decrease** in price from £10 to £5 leads to a movement **down** the demand curve from point A to point C
  - Due to the **decrease** in price, the QD has **increased** from 10 to 15 units

Your notes

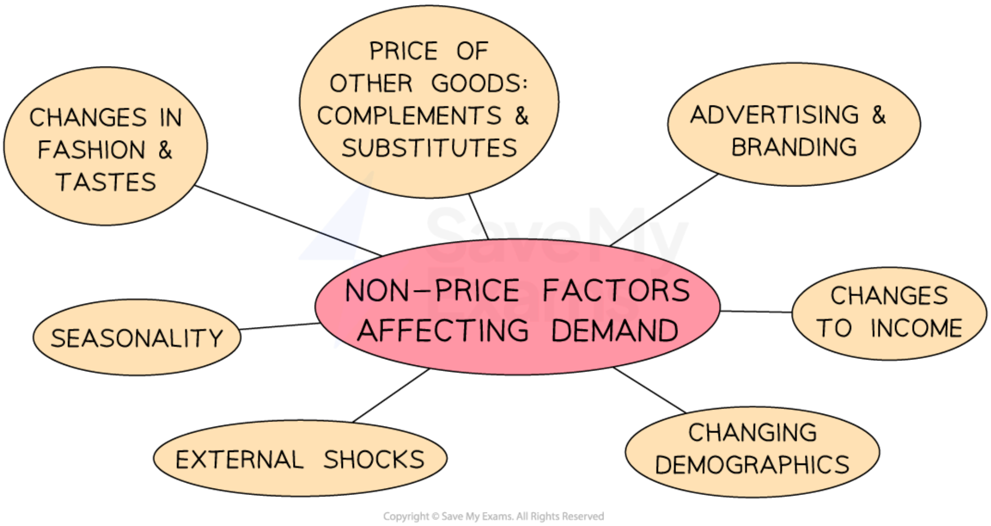
Examiner Tips and Tricks

When writing about a **movement along** the demand curve, we use the term "**quantity demanded**".

Factors leading to a change in demand

- A change in price leads to a **movement along the demand curve**
- However, a change in any other factors affecting demand will **shift the entire demand curve** to the **left or right**
  - These are called "**non-price factors affecting demand**"

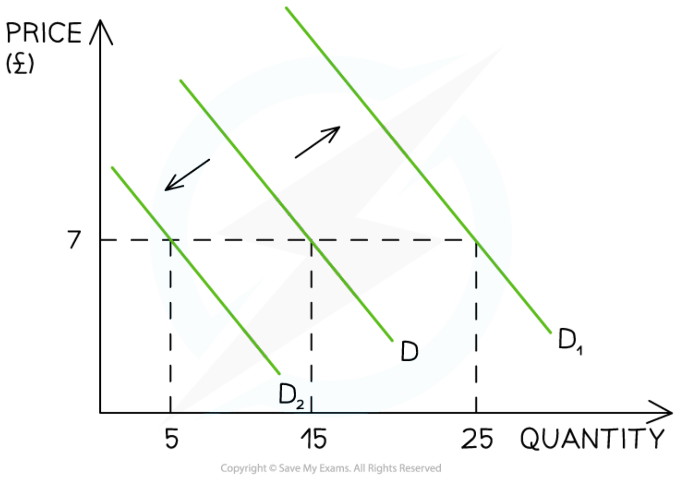
Non-price factors affecting demand



The non-price factors affecting demand result in a shift of the entire demand curve

- For example, if a firm **increases** its **Instagram advertising**, there will be an **increase in demand** as more consumers become aware of the product
  - This is a **shift in demand** from D to D_1. The price remains unchanged at £7, but the **demand has increased** from 15 to 25 units

Changes in non-price factors



A change in any non-price factor will lead to a change in the position of the demand curve

Diagram analysis

- The initial demand curve is seen at D
  - At a price of £7, 15 units are demanded
- If the **price remains constant** at £7 **but demand decreases** due to one of the non-price factors of demand (e.g. decreasing incomes), the **entire demand curve** shifts to the left, from D to D_2
  - Demand has **decreased** from 15 units to 5 units
- If the **price remains constant** at £10 **but demand increases** due to one of the non-price factors of demand (e.g. increased advertising expenditure), the **entire demand curve** shifts to the right, from D to D_1
  - Demand has **increased** from 15 units to 25 units

Non-price factors affecting demand

Non-price factor	Explanation	Example
Change in the price of substitutes	Substitute goods are **replacement goods** (e.g. different car brands)	If the **price of VW cars increases,** demand for Ford cars might rise This shifts the demand curve for **Ford cars** to the right
Change in the price of complementary goods	Complementary goods are **goods** that are **consumed together**	**Cars and petrol:** If the **price of petrol rises**, the demand for cars may fall

Your notes

		▪ This shifts the demand curve for **cars** to the left
Change in consumer incomes	▪ As a **consumer's income rises**, demand for **normal goods**) increases	▪ **Demand for branded goods** (e.g. Superdry hoodies) tends to increase as **consumer incomes rise** ▪ This shifts the demand curve for **branded goods** to the right
	▪ As a **consumer's income falls**, demand for **inferior goods** increases	▪ **Demand for own-label products** (e.g. supermarket hoodies) tends to rise as **consumer incomes fall** ▪ This shifts the demand curve for **own-label goods** to the right
Fashions, tastes and preferences	▪ If products become more fashionable, demand for them increases	▪ Plant-based foods have become **more popular** in recent years ▪ This has shifted the demand curve for **plant-based foods** to the right
Advertising and branding	▪ If more money is spent on advertising or branding, then this increases consumer awareness and brand loyalty	▪ **Coca-Cola** spends an average of $4bn per year on advertising and branding ▪ This shifts Coca-Cola's demand curve to the right and makes demand more **price inelastic**
Demographics	▪ If the **structure or size of a country's population** changes, then the demand for products will also change	▪ **A decrease in the number of babies being born** will reduce the demand for baby products ▪ This shifts the demand curve for **baby products** to the left

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Seasonality	▪ Demand varies at different **times of the year**	▪ In cold climates, the demand for **gas and electricity** falls in the summer months ▪ This shifts the demand curve for **energy** to the left
External shocks	▪ An **unexpected event** can change the demand	▪ The outbreak of **Covid-19** led to panic buying of goods such as toilet rolls ▪ This shifted the demand curve for **toilet rolls** to the right

Examiner Tips and Tricks

Remember, a change in any non-price factor that leads to **less demand** will **shift the entire demand curve to the left**, but a change in any non-price factor that leads to **more demand** will **shift the entire demand curve to the right**.

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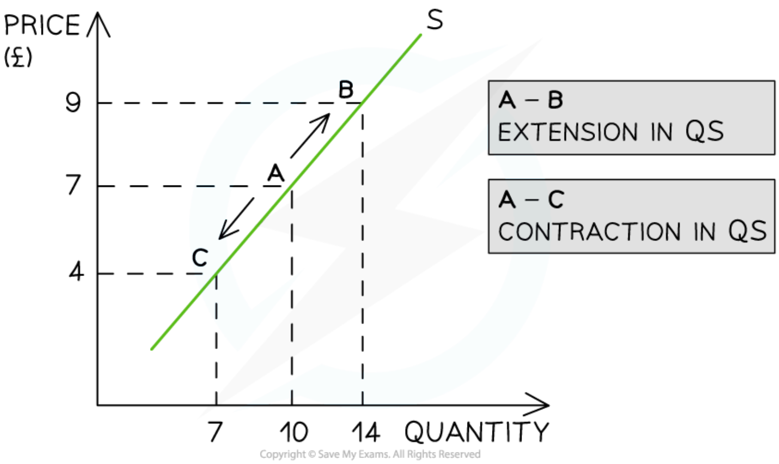
An introduction to supply

- Supply is the **number of goods/services businesses are willing to sell** at a given price in a specific time period
- There is a direct **relationship between supply and price**
  - As the **price increases**, the quantity supplied increases
  - As the **price decreases**, the quantity supplied decreases
  - At higher prices, businesses are **incentivised** to supply more of the product
  - Hence, the supply curve **slopes upwards** from left to right

A simple supply curve



A supply curve showing how a change in price will lead to a change in the quantity supplied (QS)

Diagram analysis

- An **increase** in price from £7 to £9 leads to a move **up** the supply curve, from point A to point B
  - Due to the **increase** in price, the quantity supplied (QS) has **increased** from 10 to 14 units
- A **decrease** in price from £10 to £7 leads to a movement **down** the supply curve, from point A to point C
  - **Due to the decrease in price, the QS has decreased from 10 to 7 units**

Your notes

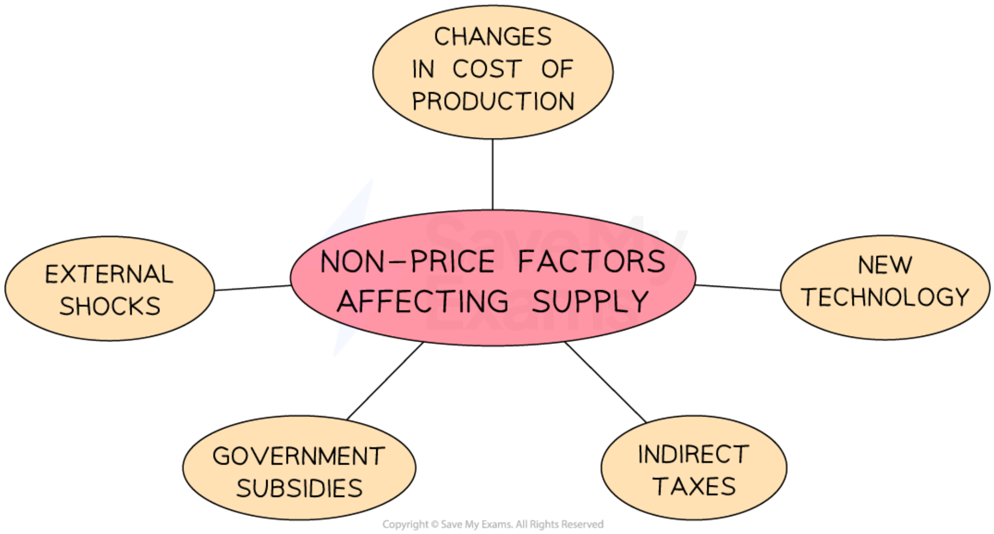
Examiner Tips and Tricks

When writing about a **movement along the supply curve**, we use the term "**quantity supplied**".

Factors leading to a change in supply

- A change in price causes a **movement along the supply curve**
- A change in any other factor affecting supply will **shift the entire supply curve** to the left or right
  - These are called **non-price factors affecting supply**

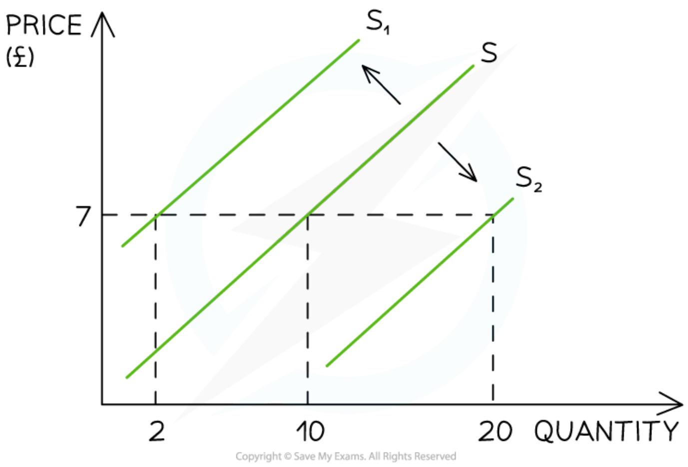
Non-price factors affecting supply



Non-price factors include changes in the costs of production, external shocks and indirect taxes

- For example, if a firm's **cost of production** increases due to an **increase in the price of a key resource**, then there will be a decrease in supply, as the firm can now only afford to produce fewer products
- This causes a **shift in supply** from S to S_1. The price remains unchanged at £7, but the **supply has decreased** from 10 to 2 units

Changes in non-price factors



A diagram showing how a change in any non-price factor of supply will shift the entire supply curve left or right

Diagram analysis

- The initial supply curve is seen at S
  - At a price of £7, 10 units are supplied
- If the **price remains constant** at £7 **but supply decreases** due to one of the non-price factors of supply (e.g. workers' wages increase), the **entire supply curve** will shift to the left, from S to S_1
  - Supply has **decreased** from 10 to 2 units
- If the **price remains constant** at £7 **but supply increases** due to one of the non-price factors of supply (e.g. cost of production falls), the **entire supply curve** will shift to the right, from S to S_2
  - Supply has **increased** from 10 to 20 units

Non-price factors affecting supply

Non-price factor	Explanation	Examples
Change in the costs of production	▪ An **increase in the cost of production** makes it **more expensive to produce** each unit, so a business will be able to produce fewer units at a given price	▪ For a **clothing manufacturer**, an increase in energy and labour costs will increase the cost of making each item ▪ This shifts the supply curve for **clothing** to the left

New technology	▪ **Advances in technology** will **lead to lower costs of production**, and businesses will be able to produce more units at a given price	▪ **Robots** have replaced many workers in **car factories**, increasing **productivity** ▪ This shifts the supply curve for **cars** to the **right**
Indirect taxes	▪ The government increases indirect taxes on businesses, which causes an **increase in the cost of production**, as firms have to pay extra costs	▪ The rate of **VAT increased from 17.5% to 20%** in the UK in 2011 ▪ This shifted the supply curves **for all businesses to the left**
Government subsidies	▪ A **producer subsidy** given by the government to businesses **will reduce the cost of production**	▪ A subsidy to the largest battery producer in the UK lowers the cost of production for **electric car manufacturers** ▪ This has shifted the supply curve for **electric vehicles** to the **right**
External shocks	▪ An **unexpected event** can change supply	▪ The **outbreak of Covid-19** caused many hotels, bars and restaurants to close down ▪ This shifted the supply curves for **hotels**, **bars and restaurants** to the **left**

Your notes

Examiner Tips and Tricks

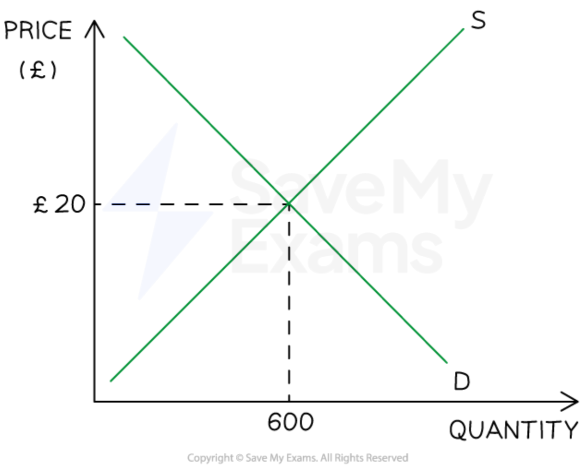
Remember, a change in any factor that leads to less supply will shift the supply curve to the left, but a change in any factor that leads to more supply will shift the supply curve to the right.

Your notes

The interaction of supply and demand

- In a market, prices for goods/services are determined by the **interaction of demand and supply**
- A **market** is any place that brings **buyers** and **sellers** together to trade at an agreed price
  - Markets can be **physical** (e.g. McDonald's) or **virtual** (e.g. eBay)
  - Buyers agree on the price **by purchasing** the good/service
  - If they do not agree on the price, then they **do not purchase** the good/service
- Based on this interaction with buyers, **sellers** will gradually **adjust their prices** until there is an **equilibrium price** and **quantity** that works for both parties
  - At the equilibrium price, **sellers** will be satisfied with the **rate/quantity** of sales
  - At the equilibrium price, **buyers are satisfied** that the product provides benefits worth paying for

Equilibrium



A graph showing a market in equilibrium with a market-clearing price of £20 and quantity of 600 units

Diagram analysis

- If the price is set at £20, demand will equal supply, and an **equilibrium** is reached
  - 600 units will be demanded and supplied
  - If the price were set above the equilibrium, **supply would be greater** than demand, and there would be a surplus

- If the price were set below the equilibrium, **demand would be greater** than supply, and there would be a shortage

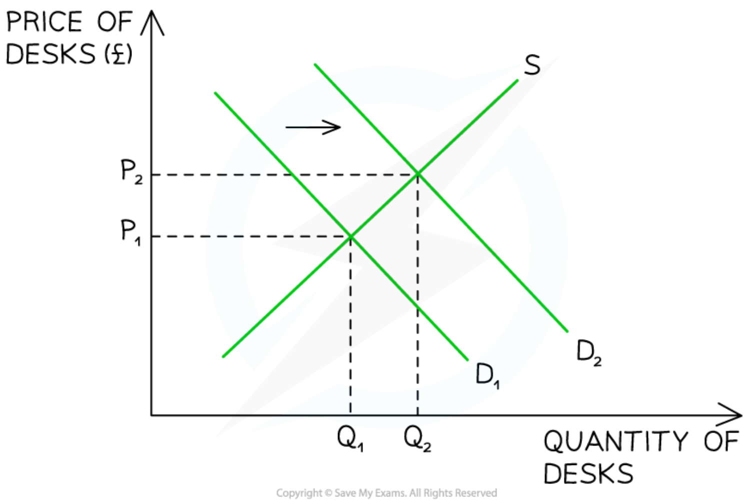
Your notes

Dynamic changes in markets

- There are **four diagrams** that can be used to show the **causes and consequences** of changes to the non-price factors of demand and supply

The impact of changes to the non-price factors of demand and supply

A rise in demand

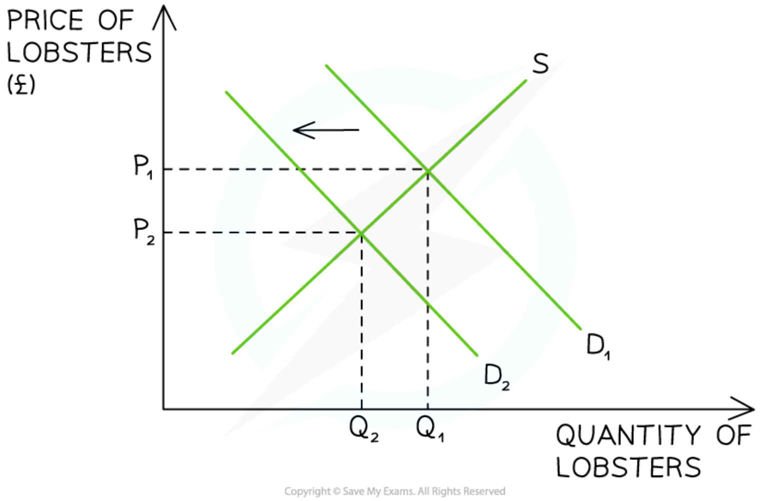


Supply and demand graph for desks, showing a rightward shift in demand from D1 to D2

Explanation

- The original equilibrium was at P_1 and Q_1
- A rise in demand causes the demand curve to **shift to the right** from $D_1 \rightarrow D_2$ (perhaps due to an increase in working from home)
- At the **original price** of P1, there is **now a shortage**, as demand exceeds the supply
- The shortage causes prices to rise from P_1 to P_2
- A **new equilibrium** develops at a price of P_2 and a quantity of Q_2 units
- The business revenue (P × Q) has changed from P_1Q_1 to P_2Q_2

A fall in demand

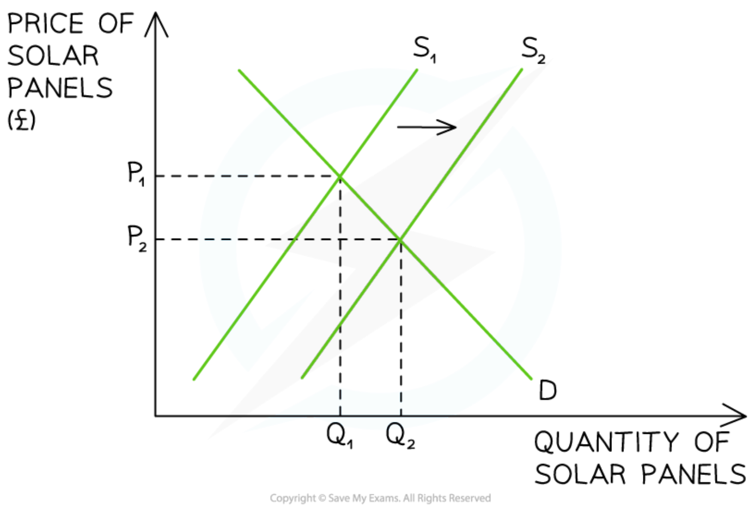


Graph showing supply and demand for lobsters, with a leftward demand shift from D1 to D2

Explanation

- The original equilibrium was at P_1 and Q_1
- A fall in demand causes the demand curve to **shift to the left** from $D_1 \rightarrow D_2$ (perhaps due to an external shock)
- At the **original price** of P1, there is **now a surplus**, as supply exceeds demand
- The surplus causes prices to fall from P_1 to P_2
- A **new equilibrium** develops at a price of P_2 and a quantity of Q_2 units
- The business revenue (P × Q) has changed from P_1Q_1 to P_2Q_2

A rise in supply



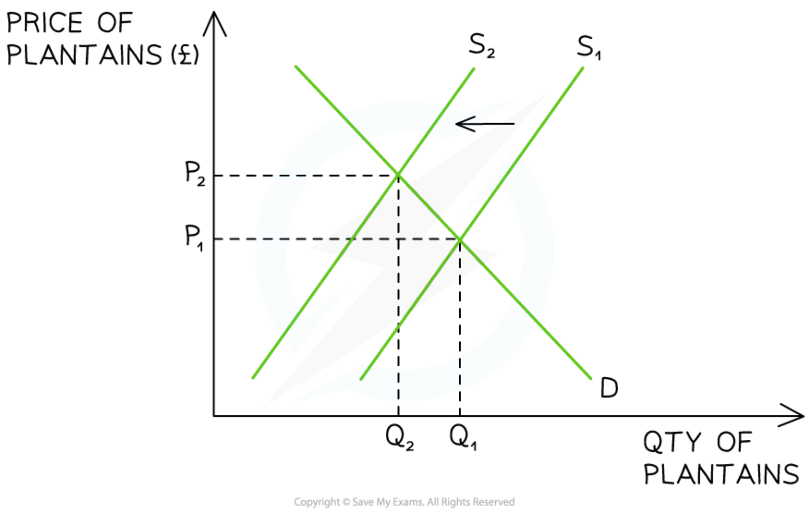
Supply and demand graph for solar panels, showing a shift from supply curve S1 to S2

Your notes

Explanation

- The original equilibrium was at P_1 and Q_1
- A rise in supply causes the supply curve to **shift to the right** from $S_1 \rightarrow S_2$ (perhaps due to an increase in productivity)
- At the **original price** of P_1, there is **now a surplus**, as supply exceeds demand
- The surplus causes prices to fall from P_1 to P_2
- A **new equilibrium** develops at a price of P_2 and a quantity of Q_2 units
- The business revenue (P × Q) has changed from P_1Q_1 to P_2Q_2

A fall in supply



Supply and demand graph for plantains, showing S1 shifting left to S2

Explanation

- The original equilibrium was at P_1 and Q_1
- A fall in supply causes the supply curve to **shift to the left** from $S_1 \rightarrow S_2$ (perhaps due to an increase in the costs of production)
- At the **original price** of P_1, there is **now a shortage**, as demand exceeds supply
- The shortage causes prices to rise from P_1 to P_2
- A **new equilibrium** develops at a price of P_2 and a quantity of Q_2 units
- The business revenue (P × Q) has changed from P_1Q_1 to P_2Q_2

Examiner Tips and Tricks

Your notes

When answering questions on changes to markets, remember that a change in the price of the good leads to a movement along the demand or supply curve, not a shift in the demand or supply curve.

Practise drawing and labelling demand and supply diagrams accurately, describing them step by step. This will help you gain analysis marks. You can also use demand and supply diagrams to illustrate changes in total revenue.

Your notes

An introduction to price elasticity of demand

- When there is an **increase in price**, there will be a **fall in the quantity demanded (QD)**, and when there is a **fall in price**, there will be an **increase in the QD**
- The question businesses are interested in is "**By how much** will **the QD change?**"
- The **price elasticity of demand** (PED) helps us to calculate **how responsive the change in QD will be to a change in price**
  - **The responsiveness is different for different types of products**

Calculation of PED

- **The PED value is always negative**
- **PED can be calculated** using the following formula:

$$\text{PED} = \frac{\%\ \text{change in quantity demanded}}{\%\ \text{change in price}} = \frac{\%\Delta\ \text{in QD}}{\%\Delta\ \text{in P}}$$

- **To calculate a % change**, use the following formula:

$$\%\ \text{Change} = \frac{\text{new value} - \text{old value}}{\text{old value}} \times 100$$

Worked Example

The PED for popcorn at the cinema is -0.8. The current price of a box of popcorn is £5. Using the data, calculate the percentage change in QD following a £1 increase in the price of a box of popcorn. You are advised to show your work.

[4]

Step 1: State the PED formula:

$$\text{PED} = \frac{\%\ \text{change in demand}}{\%\ \text{change in price}}$$ **[1]**

Step 2: Calculate the percentage change in price:

$$= \frac{£6 - £5}{£5} \times 100$$ **[1]**

$$= 20\%$$

Step 3: Insert the data you have been given into the formula:

$$-0.8 = \frac{x}{20\%} \text{ [1]}$$

Step 4: Rearrange and solve for x:

$$x = -0.8 \times 20$$

[1]

$$x = -16\%$$

Examiner Tips and Tricks

Remember, **if the price decreases, QD increases**, and **if the price increases, QD decreases**. In this case, price increases. Therefore, QD must fall.

Interpretation of PED numerical values

- The numerical value of PED indicates the **responsiveness** of a change in QD to a change in price
- PED will always be negative due to the **inverse relationship** between price and quantity
  - If the price goes up, the QD goes down
  - If the price goes down, the QD goes up

Interpretation of PED values

Numerical value	Type of good	Explanation
-1 or lower	**Elastic** ▪ Examples include **luxury products** such as cars, smart watches, foreign holidays, cinema visits, jewellery and branded goods	▪ Demand is **more responsive to a change in price** ▪ The %Δ in QD is **more than proportional** to the %Δ in P
Between 0 and -1	**Inelastic** ▪ Examples include **necessities** such as bread, milk, eggs, potatoes, fuel, rent and toothpaste ▪ **Addictive products** such as cigarettes and sugary foods	▪ Demand is **less responsive to a change in price** ▪ The %Δ in QD is **less than proportional** to the %Δ in P

Your notes

Your notes

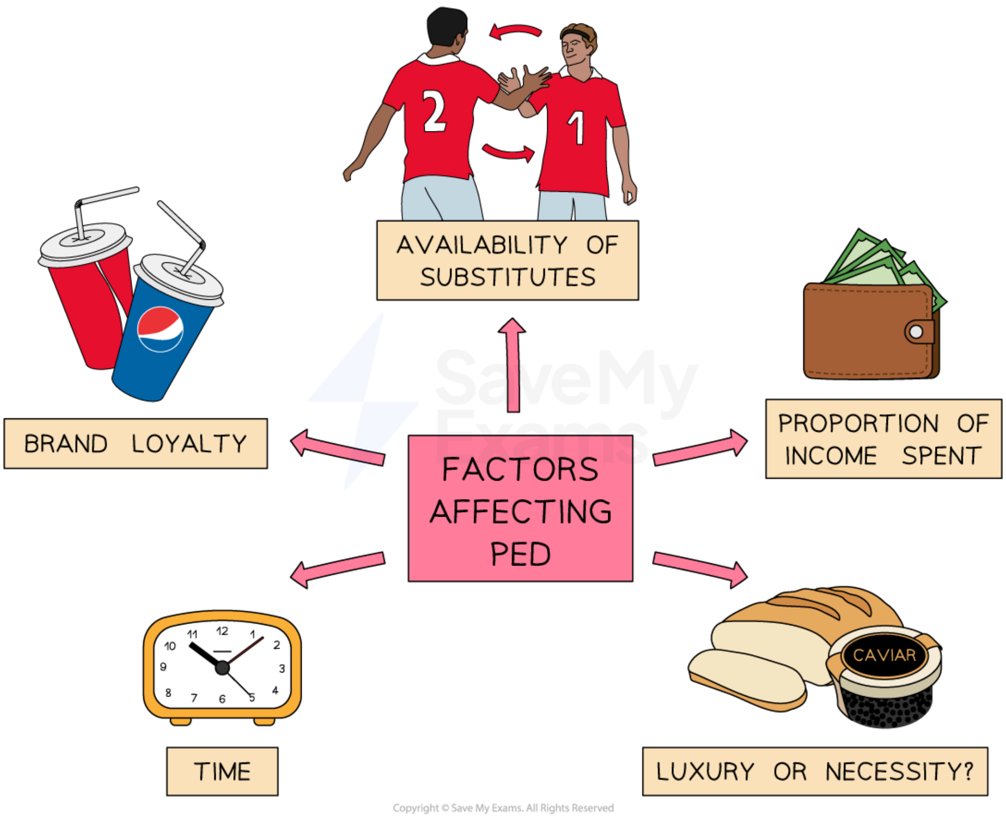
Examiner Tips and Tricks

Students often confuse the negative sign in the answer to PED questions. Do not assume that the negative is mathematical, such that an elasticity of –1 is smaller than –0.3. It is larger (more price elastic).

The **PED will always be negative, indicating the inverse relationship between price and QD** (i.e. when the price increases, QD decreases, and when the price decreases, QD increases).

When interpreting the value of PED, do not say that "the product is elastic or inelastic"; it is better to say that "**demand** for the product is price elastic or price inelastic".

Factors influencing the PED



The factors that determine whether the demand for a product is more price elastic or price inelastic

Brand loyalty

- The aim of advertising and marketing expenditure by a business is to shift the demand curve to the right and **make the demand more price inelastic**

- For example, Coke consumers are more brand loyal to Coca-Cola and refuse to buy Pepsi, even though their taste is very similar

Availability of substitutes

- PED will be more price inelastic (lower) for goods that have **fewer substitutes**
  - For example, petrol has fewer substitutes and is more price inelastic, whereas chocolate bars have more substitutes and are more price elastic

The proportion of income taken up by the product

- The **smaller the proportion of income** we spend on a product, the **more price inelastic** the demand will be
  - For example, a small amount of income is spent on salt, so the demand for salt will be more price inelastic. However, buying a new car takes up a **bigger proportion** of consumer income, so **the PED is higher**

Luxury or necessity

- **Necessities** are required as part of consumers' daily needs, and therefore, demand for them is **more price inelastic**
  - For example, bread, milk, petrol, gas and electricity might be considered necessities
- **Luxuries** are not essential, and therefore, demand for them is **more price elastic**
  - For example, smoked salmon, Nike Air Jordans and foreign holidays might be considered luxuries

Time

- The **longer the time period** under consideration, the more price elastic the demand for a good or service is likely to be (consumers have more time to search for substitutes)
- The **shorter the time period** under consideration, the more price inelastic the demand for a good or service is likely to be
  - For example, if the price of petrol increases, making driving more expensive, there is little that consumers can do in the short term. However, they may switch to alternatives such as public transport or bicycles in the long term

The significance of PED to businesses

- If businesses can **determine the PED** for their products, they can adjust their **pricing strategy** to **maximise their revenue**
- If demand for their products is relatively **price inelastic (PED < -1)**, raising the price will lead to an **increase in total revenue**. However, lowering the price will lead to a **fall in total revenue**
  - **Price skimming strategies** are best employed for products that are price inelastic in demand

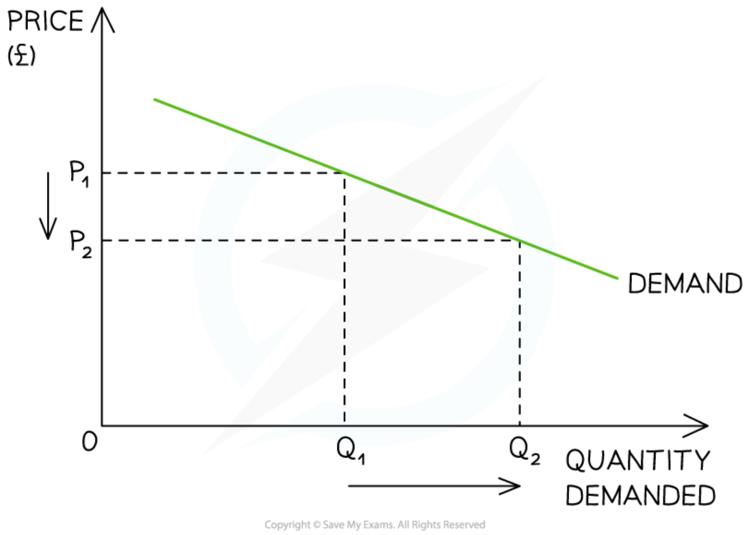
Your notes

- If demand for their products is relatively **price elastic (PED >-1)**, raising the price will lead to a **fall in total revenue**. However, lowering the price will lead to a **rise in total revenue**
    - **Competitive pricing strategies** are best employed for products that are price elastic in demand

Your notes

The relationship between PED and total revenue

Price elastic demand

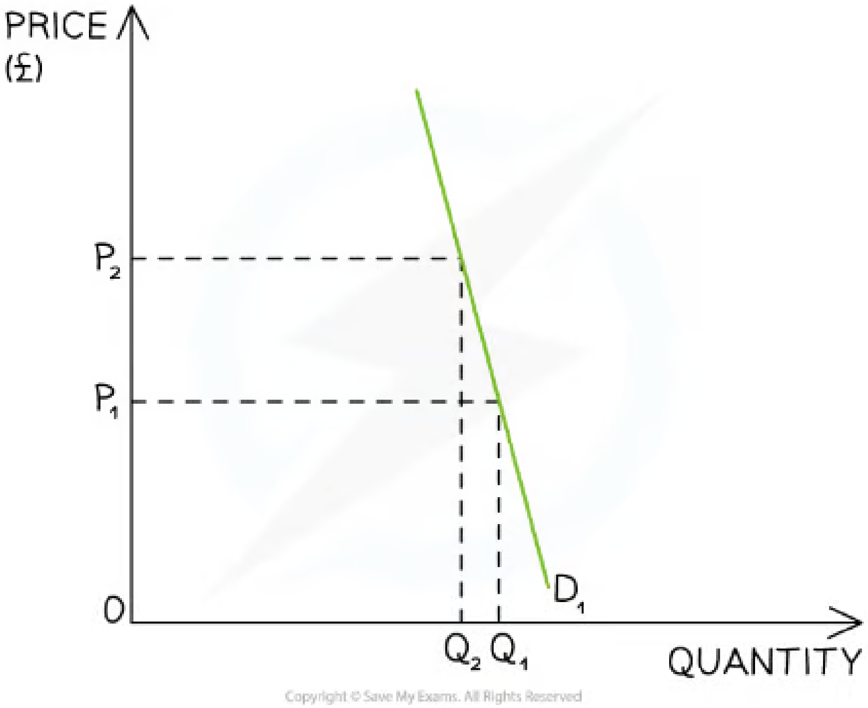


Price elastic demand

- PED is **greater than 1**
- An increase in the selling price reduces the total amount of revenue generated from sales
- A reduction in the selling price increases the total amount of revenue generated from sales

Price inelastic demand

Your notes



Price inelastic demand

- PED is **between 0 and 1**
- An increase in the selling price increases the total amount of revenue generated from sales
- A reduction in the selling price reduces the total amount of revenue generated from sales

Definition and calculation of income elasticity of demand

- **Changes in income** result in changes to the **demand** for products
  - Businesses are interested **in how much** the **quantity demanded (QD) will change** for different products
- The **income elasticity of demand (YED)** reveals how **responsive** the change in **QD** is to a change in **income**

Calculation

- YED can be calculated using the following formula:

$$\text{YED} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in income}}$$

Worked Example

An individual's income falls from £450 per week to £405 per week. As a result, their demand for takeaway meals falls from five per week to three per week.

Calculate the YED for takeaway meals.

[4]

Step 1: State the YED formula:

$$\text{YED} = \frac{\%\text{ change in QD}}{\%\text{ change in Y}}$$ [1]

Step 2: Calculate the % change in QD:

$$\%\text{ change in QD} = \frac{5 - 3}{5} \times 100$$ [1]

$$= -40\%$$

Step 3: Calculate the % change in Y:

$$\%\text{ change in Y} = \frac{£450 - £405}{£450}$$ [1]

$$= -10\%$$

Step 4: Insert the above values into the YED formula:

$$YED = \frac{-40\%}{-10\%}$$ [1]

$$= 4$$

Your notes

Interpretation of numerical YED values

- The YED value can be **positive or negative**, and the value is important in determining the type of good
  - A good with a **positive YED** value is considered to be a **normal good**
    - Normal goods can be classified as **necessities** or **luxuries**
  - A good with a **negative YED** value is considered to be an **inferior good**

Interpretation of the numerical values of YED

Numerical value	Type of good	Explanation
>1	**Luxury** ▪ Examples include cars, smart watches, foreign holidays, cinema visits, jewellery and branded goods	▪ Demand rises when income rises, and demand falls when income falls ▪ Demand is responsive to a change in income (income elastic)
0–1	**Necessity** ▪ Examples include staple food items, such as bread, milk, eggs and potatoes, as well as fuel and toothpaste	▪ Demand is not very responsive to a change in income (income inelastic)
<0	**Inferior** ▪ Examples include public transport, domestic holidays, canned foods and unbranded/own-label goods	▪ Demand rises when income falls (negative income elasticity) ▪ Demand falls when income rises

The factors influencing YED

- YED is influenced by **many factors in an economy** that change the wages of workers

- During a **recession,** wages sometimes fall and, more likely, do not rise significantly
  - Demand for **inferior goods** rises while demand for **luxury goods** falls
- During a period of **economic growth** and rising wages, demand for luxury goods increases while demand for inferior goods decreases
- Other influences on income include **minimum wage legislation**, **taxation** and **increased international trade**

- YED is also influenced by the **nature of the good**, as discussed above
  - Luxury or necessity (both are classified as normal goods)
  - **Inferior** or **normal** good

The significance of YED to businesses

- Understanding the YED is useful to businesses, as it can **help** them **plan their production and products**
  - Planning in this way will help them generate higher profits and have less exposure to downturns in the economy

Production planning

- A business needs to **plan how much it is going to produce**, which will help it **determine the number of resources**, such as raw materials and labour, **it will need**
- If a business can determine the YED for its products and can accurately predict changes in income, then it can **plan whether to increase or decrease production**
- It can help managers with financial planning
- Production planning is **easier when the YED is relatively inelastic**, as demand is likely to be more constant

Product planning

- The **economy goes through different stages over time**, from recession to recovery and growth, and so **incomes** will **fluctuate**
- This is known as the **business cycle**
- During a **recession**, producers of **inferior goods** will benefit from **higher demand** but will lose out when incomes rise and consumers return to normal goods
- Some businesses might have different products in their **product portfolio** to take account of this. For example:
  - Tesco has its "Finest", "Standard" and "Value" range to appeal to all income segments of the market
  - During the 2008 recession, Waitrose introduced its "Essentials" range of products to appeal to more budget-conscious shoppers
  - VW owns Skoda, Audi and Porsche, and it has different products within its range to appeal to different income groups

Your notes